Deliverable 2: Architectural Justification Document (Max 1 Page)

Architecture Selection

We chose **Layered (3-Tier) Architecture** for the **Peer-to-Peer Carpooling & Expense Splitter System**.

1. Two Scenario-Related Reasons for Choice

- **Isolation of Complex Geospatial Calculations:** The system requires real-time route matching (FR-001) while strictly enforcing a maximum 1.5 km detour tolerance limit for Driver Hosts (FR-002). Layered architecture separates client presentation from backend processing, ensuring heavy spatial trajectory calculations occur within a dedicated Business Logic tier without locking or slowing down mobile client interfaces.
- **Decoupling Financial & Expense Calculation Algorithms:** The system automatically calculates and splits fuel costs among all confirmed trip participants (FR-005). Isolating payment logic in a dedicated business layer allows updates to cost-splitting formulas (e.g., dynamic pricing or fuel efficiency parameters) without modifying trip creation (FR-003) or search interfaces (FR-004).

2. Security Advantage

- **Tier Boundary Enforcement & Encrypted Data Protection:** In accordance with **NFR-001**, user ratings, driver license verification documents, and travel histories must be stored securely with encrypted backups. A Layered Architecture prevents client applications from communicating directly with the database. All queries pass through authenticated API controllers in the Business Layer. Sensitive data (such as driver license images) is encrypted at rest using AES-256 before reaching the Database Component, ensuring unauthorized access is blocked at the system boundary.

3. Performance Benefit

- **Achieving the Sub-3-Second Response SLA (NFR-002):** **NFR-002** specifies that route matching and expense calculation results must be delivered within **3 seconds** under both normal and peak operating conditions. Layered architecture supports database connection pooling and spatial route caching (e.g., Redis spatial indexing) at the business logic layer. This minimizes database roundtrips for common commute routes, allowing the system to compute compatibility and cost splits within the required 3-second window.